



IUCN SCHOLARLY PROGRAM – BAHARI YETU PROJECT

TRAINING NEEDS ASSESSMENT (TNA) REPORT

Data Management & Analysis in R for Environmental and Climate Sciences

A comprehensive evaluation of programming baseline familiarity, tool utilization, and core analytical competencies of the sponsored PhD and MSc cohort across East African universities.

Target Audience: PhD & MSc Scholarship Beneficiaries
Participating Institutions: SUA, UDSM, NM-AIST, SUZA
Sponsor: International Union for Conservation of Nature (IUCN)
Project Framework: Bahari Yetu Project

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Executive Summary

This Training Needs Assessment (TNA) Report presents a comprehensive evaluation of the data management, visualization, and analysis capabilities of the PhD and MSc students sponsored by the **International Union for Conservation of Nature (IUCN)** under the **Bahari Yetu Project**.

A structured questionnaire was distributed to all 28 sponsored beneficiaries to identify the gap between their current skills and the competencies required for their postgraduate research. We received **27 completed responses** (a 96.4% response rate), representing four East African universities.

Key findings indicate:

- **Severe R Skill Gaps:** 48.1% of the students have never written code in R, and another 29.6% can only run R scripts written by others.
- **Low Familiarity with Modern Workflows:** 74.1% are unfamiliar with “Tidy Data” principles, and 85.2% have never used version control (Git/GitHub).
- **High Interest in Learning:** Students expressed an overwhelming desire to acquire these skills (average interest score of 4.59 out of 5), showing critical needs in Advanced Statistical Modeling, Reproducible Reporting (Quarto/Markdown), and Spatial Data GIS Analysis.

Based on these findings, we propose a customized **one-week intensive training course** utilizing a **“Bring Your Own Data” (BYOD)** delivery model. This will ensure that students apply the lessons directly to their active thesis research datasets.

Acknowledgements

We express our sincere gratitude to the **International Union for Conservation of Nature (IUCN)** for their generous funding and comprehensive support under the framework of the **Bahari Yetu Project**. This initiative plays a vital role in addressing scientific knowledge gaps and building institutional research capacity within East African academic and environmental sectors.

Special thanks are extended to the participating institutions:

- **Sokoine University of Agriculture (SUA)**
- **University of Dar es Salaam (UDSM)**
- **Nelson Mandela African Institution of Science and Technology (NM-AIST)**
- **State University of Zanzibar (SUZA)**

We are also deeply grateful to the MSc and PhD scholarship beneficiaries who took the time to participate in the survey, providing the critical insights that helped shape this Training Needs Assessment. Finally, we thank the project coordinators, scholarly leads, and facilitators for their dedication to organizing and designing this capacity-building program.

1 Cohort Genesis & Profile

The academic scholarship program, funded under the **IUCN Bahari Yetu Project**, sponsors 28 students from four major universities in East Africa. The cohort is highly structured, bridging diverse research disciplines in environmental, agricultural, and marine sciences.

The primary mandate of the Bahari Yetu Project is to conserve, manage, and sustainably utilize coastal and marine ecosystems in Tanzania. By funding these scholarships, IUCN aims to address critical scientific knowledge gaps in local environmental protection, resource management, and climate change reporting. Sponsoring 21 MSc and 7 PhD students across top-tier East African universities creates a robust cohort of next-generation researchers equipped to tackle complex regional challenges, such as forest biodiversity loss, industrial and agricultural greenhouse gas (GHG) emissions, marine connectivity, and macroplastic pollution.

1.1 Demographic & Institutional Distribution

Table 1 summarizes the institutional and degree level distribution of the 27 students who responded to the survey. The cohort includes:

- **Gender Breakdown:** 16 Female students and 11 Male students, demonstrating a strong emphasis on gender inclusion in environmental sciences.
- **Academic Level:** 20 MSc candidates and 7 PhD candidates, providing a healthy mix of early-career and advanced researchers.

Table 1: Institutional and Degree Level Distribution of the TNA Respondents

University	MSc	PhD	Total
NMAIST	3	2	5
SUA	12	2	14
SUZA	1	0	1
UDSM	4	3	7
Total	20	7	27

The institutional enrollment reflects targeted capacity building illustrate in Table 2:

1. **Sokoine University of Agriculture (SUA):** Houses the largest segment of the cohort (14 candidates). This figure aligns with SUA's specialized laboratory infrastructure and research focus on forestry, agricultural GHG inventories (AFOLU), and waste-to-energy technologies.

2. **University of Dar es Salaam (UDSM)**: Represents 7 candidates, heavily engaged in marine biology, marine genetics, and plastic waste transport modeling in river catchments.
3. **Nelson Mandela African Institution of Science and Technology (NM-AIST)**: Has 5 candidates, bringing computational strength to industrial emissions modeling and ML-based waste predictions.
4. **State University of Zanzibar (SUZA)**: Has 1 candidate, focused on local marine ecology and island fisheries.

Table 2: Institutional and Degree Level Distribution of the candidates by research themes

University	Degree	Number	Research Themes
NMAIST	MSc	3	Coastal Forest; Waste; Coastal Forest
NMAIST	PhD	2	Waste; Ippu
SUA	MSc	12	Energy; Ippu; Energy; Afolu; Afolu; Ippu; Coastal Forest; Marine Ecology; Plastic Waste; Energy; Energy; Ippu
SUA	PhD	2	Coastal Forest; Energy
SUZA	MSc	1	Plastic Waste
UDSM	MSc	4	Plastic Waste; Waste; Waste; Marine Ecology
UDSM	PhD	3	Plastic Waste; Plastic Waste; Marine Ecology

Figure 1 visually illustrates this distribution, showing the cross-institutional breakdown of MSc and PhD candidates, faceted by gender.

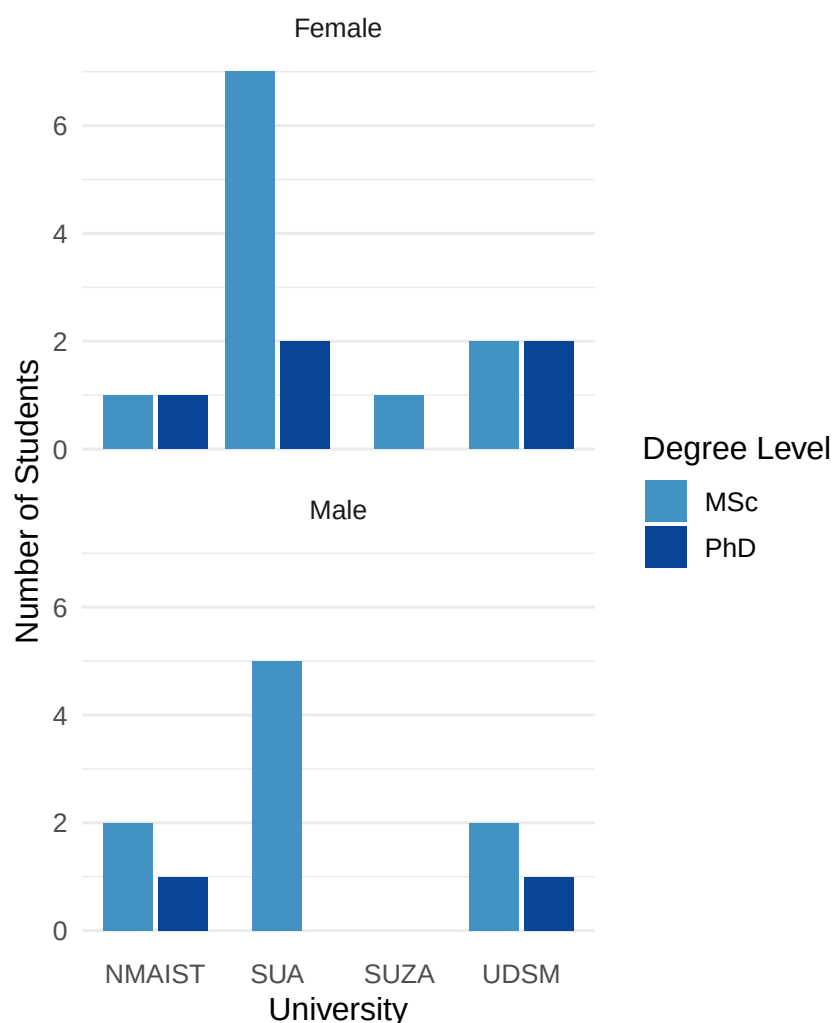


Figure 1: Cohort Distribution by University, Degree Level, and Gender

1.2 Research Themes

The research themes are designed symmetrically to match national climate goals and ecosystem assessments. The cohort is split across seven distinct environmental research themes illustrated in Figure 2. These themes are:

1. **Coastal Forest:** Focuses on woody species status, community-based forest management (CBFM) village forests, molecular distinctiveness, and phytochemical screening. These studies provide empirical baselines for terrestrial carbon storage and forest biodiversity conservation in coastal reserves.
2. **GHG - Industrial Processes and Product Use (IPPU):** Investigates industrial greenhouse gas emissions, cement process emissions, and uncertainty estimation. This research directly supports Tanzania's compliance with national GHG inventory standards.

3. **GHG - Energy:** Studies biomass energy use, non-renewable biomass fractions, carbon accounting, and habitat impacts. It addresses household energy transitions and renewable energy strategies.
4. **Marine Ecology:** Focuses on seagrass genetics/connectivity, green turtle diets, dolphin acoustics (F-PODs), boat-based surveys, and fish biodiversity under climate pressure.
5. **GHG - Agriculture, Forestry, and Other Land Use (AFOLU):** Targets crop land activity data consistency and regional agricultural GHG inventories, which are essential for sustainable agricultural policies.
6. **Plastic Waste Management:** Focuses on spatial modeling of macroplastics, catchment transport pathways, and intervention prioritization in municipal river systems.
7. **GHG - Waste:** Investigates methane potential optimization, municipal solid waste prediction using remote sensing and machine learning, and Life Cycle Assessments (LCA) to design circular economy interventions.

Figure 2 shows how degree levels are distributed across these research themes, highlighting the balance of thematic focus across MSc and PhD students.

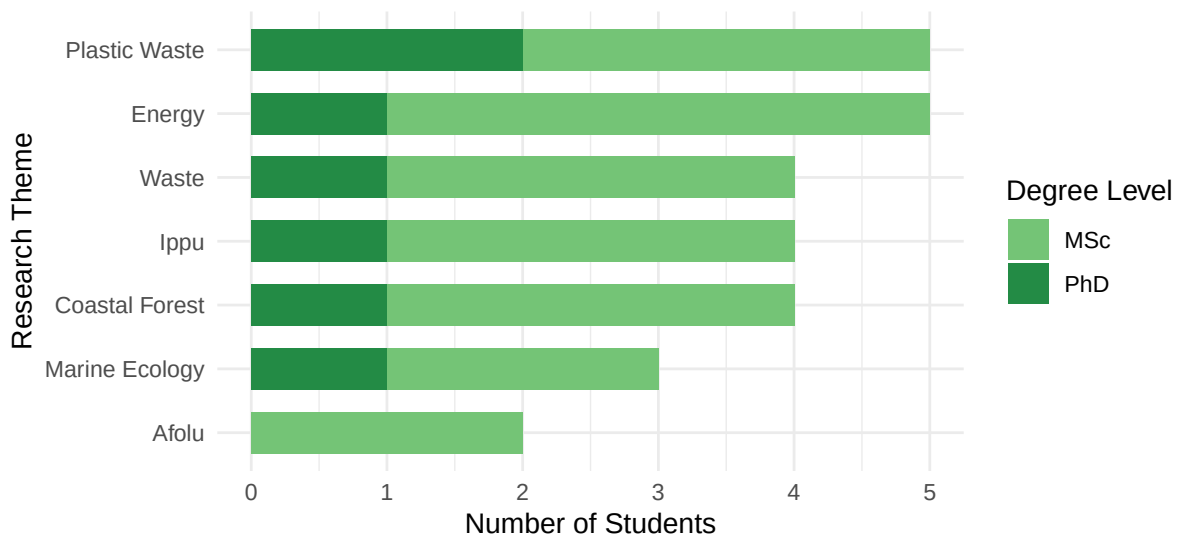


Figure 2: Academic Level Distribution Across Research Themes

2 Current Data Environment & Challenges

Understanding the tools, formats, and allocations of research time that students currently rely on is essential for designing effective translation pathways to R programming. The survey reveals a cohort that is heavily dependent on legacy tools and manual methods, resulting in significant time sinks during the data preparation phase.

2.1 Software Tools & Data Formats In Use

Currently, Microsoft Excel is the dominant tool used for data storage and analysis, while spatial analysis tools (QGIS and ArcGIS) and R are moderately used (Figure 3). Excel is used by 20 students, highlighting its ubiquitous role as both a database and analytical workbench. This creates challenges, as spreadsheets lack automated audit trails, are prone to manual entry errors, and scale poorly when handling complex datasets.

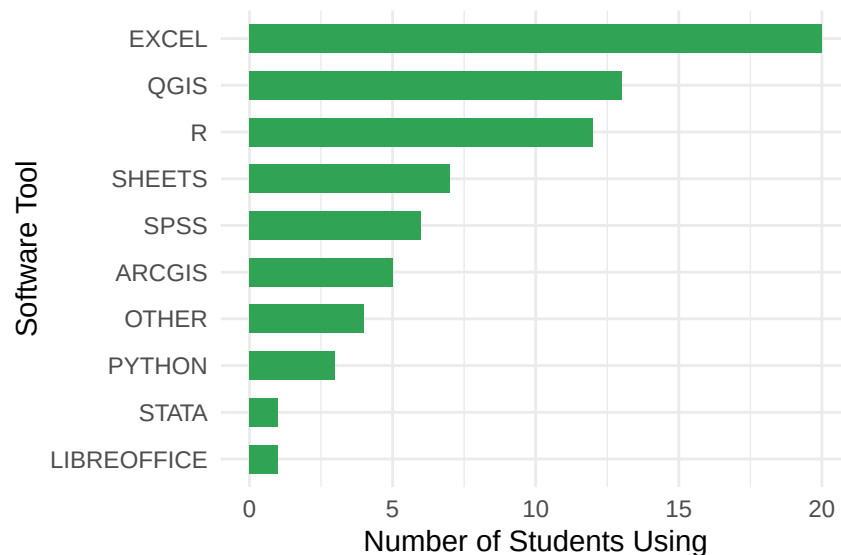


Figure 3: Primary Software Tools Currently Used by Students

The students collect diverse types of raw data, which can be grouped into several primary formats. As shown in Figure 4, tabular data and survey formats (e.g., household interviews) are the most common, followed closely by spatial coordinates and shapefiles.

- **Tabular data:** 20 students collect spreadsheets or CSV files.
- **Survey data:** 18 students compile interviews and questionnaires.
- **Spatial data:** 17 students manage GIS vectors and rasters.
- **Laboratory experimental data:** 11 students gather chemical assays or genetics counts.
- **Audio or sensor logs:** 7 students collect acoustic recordings or sensor timeseries.

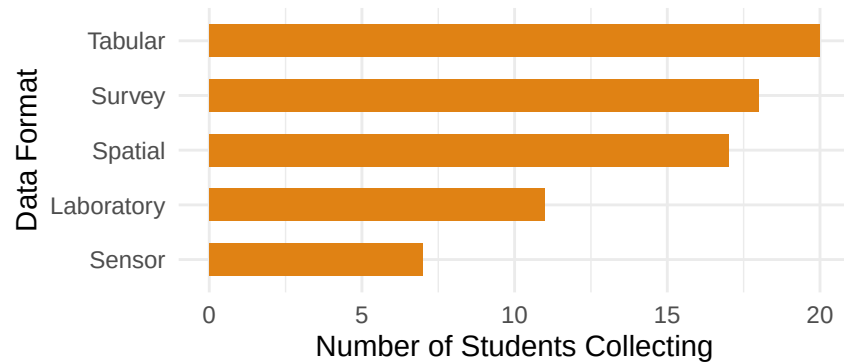


Figure 4: Primary Data Formats Collected by Students

2.2 Allocation of Research Time

A major finding of the TNA is the disproportionate amount of time students spend preparing and cleaning data rather than analyzing or reporting it. On average, students report spending **21.1%** of their total research time simply cleaning and formatting raw data. Figure 5 shows the distribution of research time allocation across the cohort. This empirical bottleneck highlights that the lack of automated wrangling tools forces researchers to spend their most valuable hours in manual spreadsheets instead of performing statistics and writing publications.

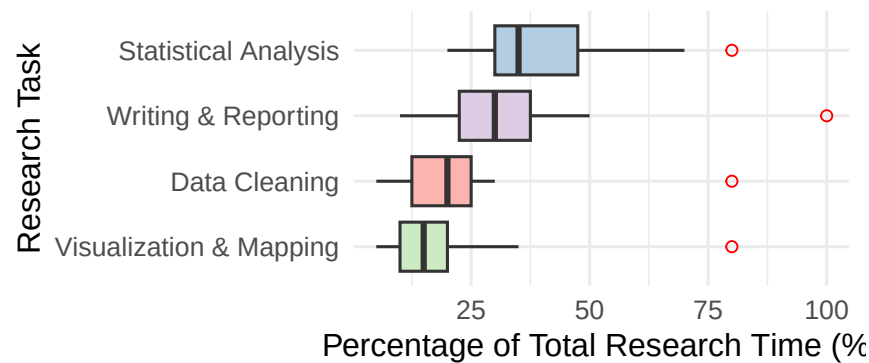


Figure 5: Distribution of Research Time Allocation

2.3 Primary Data Bottlenecks & Frustrations

When asked about their single biggest bottleneck, the students' responses fell into four major thematic areas:

1. **Data Integration & Cleaning:** Many students struggle to combine disparate datasets (such as household energy surveys and geographical forest plots) into a single tidy table.

Representative Student Quote: > “Integrating and organizing different types of data (survey, spatial, and fuelwood consumption data) while ensuring data quality and preparing them for greenhouse gas emission estimation and analysis.”

2. **Reproducibility & Coding Blocks:** Students face major friction when moving from point-and-click tools to R, often stalling due to code syntax errors or lack of clear documentation.

Representative Student Quote: > “My biggest bottleneck is analyzing experimental data efficiently and producing reproducible statistical analyses and high-quality visualizations. I hope to improve these skills through training.”

3. **Large Volumes and Loss Risks:** Managing and organizing large files (e.g., acoustic sensor outputs or raster images) while avoiding duplicate files or data loss.

Representative Student Quote: > “Managing large volumes of data from different sources, ensuring accuracy, avoiding data loss, and making the data easy to retrieve and analyze.”

4. **Methodological Uncertainty:** A lack of confidence in selecting the correct statistical model (such as regression diagnostics or multivariate analysis) for their experimental designs.

3 Skill Gap Analysis

The TNA questionnaire assessed the students’ familiarity with basic coding and modern data analysis concepts, alongside self-assessments of their current skills versus their interest in learning specific competencies. The results reveal a massive disparity: while students have very low baseline skills, they have an exceptionally high interest in mastering these modern workflows to support their research.

3.1 Baseline Programming Familiarity

Familiarity with coding, “Tidy Data” principles, and version control tools is extremely low across the cohort. Figure 6 visualizes the three baseline metrics, highlighting that the majority of students are complete beginners in scientific computing workflows.

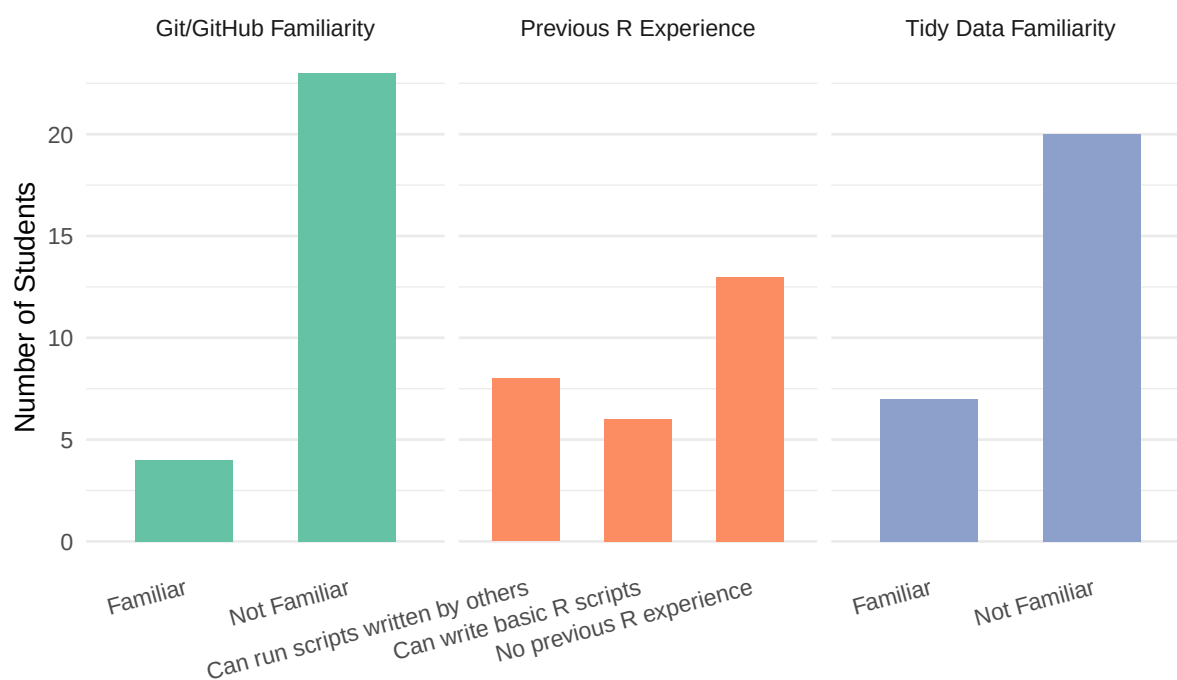


Figure 6: Baseline Programming and Workflow Familiarity

Table 3 provides the exact counts for these baseline metrics. It shows that **48.1%** of the students have never written code in R, and **85.2%** have never used version control.

Table 3: Baseline Programming and Workflow Familiarity Counts

Baseline	Students
Can run scripts written by others	8
Can write basic R scripts	6
No previous R experience	13
Familiar with Tidy Data: No	20
Familiar with Tidy Data: Yes	7
Familiar with Git/GitHub: No	23
Familiar with Git/GitHub: Yes	4

To see if students who already have basic R experience are familiar with “Tidy Data” principles, we ran a cross-tabulation (Table 4). Strikingly, even among the 6 students who can write basic R scripts, **2** are unfamiliar with “Tidy Data” principles. This indicates that even students

with some programming baseline lack exposure to reproducible data wrangling standards, confirming that the training must establish tidy principles for all participants.

Table 4: Cross-Tabulation of R Experience and Tidy Data Familiarity

r_experience	No	Yes	Total
Can run scripts written by others	6	2	8
Can write basic R scripts	2	4	6
No previous R experience	12	1	13
Total	20	7	27

3.2 Competency-Level Gap Analysis

Students rated their current skill levels and their interest in learning nine specific data management and analysis competencies on a Likert scale (1 = Lowest/None, 5 = Highest/Expert). The **Gap** is calculated as:

$$\text{Gap} = \text{Interest Score} - \text{Current Skill Score} \quad (1)$$

Table 5 displays the mean scores and the resulting gaps:

Table 5: Competency-Level Gaps (Ordered by Size of Gap)

Competency	Current skill	Interest to learn	Gap
Advanced Statistics	1.78	4.73	2.95
Reproducible Reports	1.93	4.70	2.78
Spatial Analysis	2.00	4.56	2.56
Data Merging	2.15	4.59	2.44
Data Reshaping	2.26	4.59	2.33
Data Visualization	2.37	4.58	2.21
Data Cleaning	2.48	4.56	2.07
Data Import	2.52	4.52	2.00
Basic Statistics	2.59	4.44	1.85

Competency	Current skill	Interest to learn	Gap
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Figure 7 directly compares the current skill levels and the learning interest side-by-side. While interest is uniformly high (averaging above 4.4 out of 5 in all categories), current skill levels are low, hovering between 1.7 and 2.6.

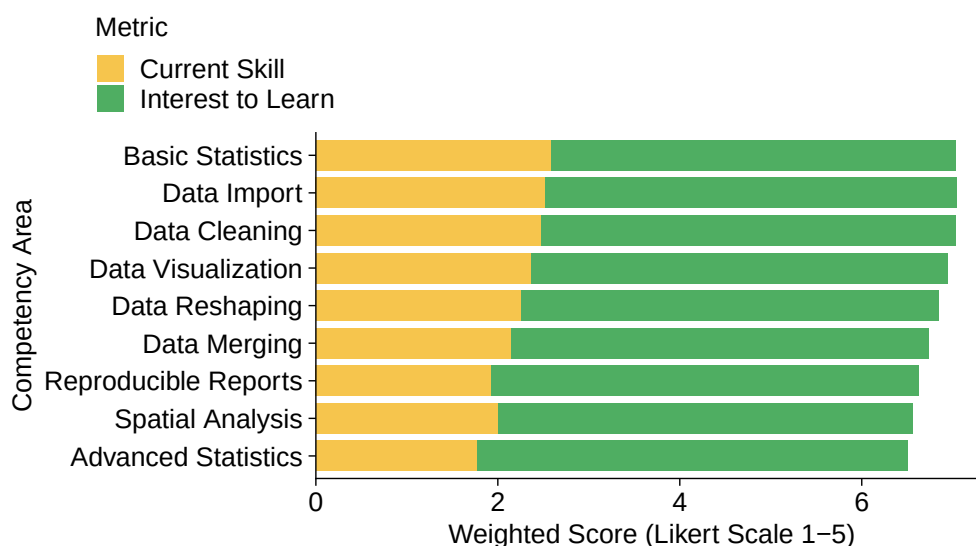


Figure 7: Comparison of Current Skill vs. Interest to Learn

The gap analysis displays the resulting net gap scores, ordered from largest to smallest (Figure 8) highlights several critical trends:

1. **Advanced Statistics & Machine Learning (Gap: 2.95):** This represents the largest overall gap. Students have almost no prior experience (mean skill: 1.78) but have the highest interest (mean interest: 4.73) due to the modeling demands of their theses (e.g., predicting landfill waste volumes, life cycle assessments, or machine learning algorithms).
2. **Reproducible Reports (Gap: 2.78):** Students currently rely on manual, repetitive copy-pasting of figures and tables from R or Excel into Word. They have high interest (4.70) in learning Quarto to automate reporting.
3. **Spatial Analysis & GIS in R (Gap: 2.56):** Reflects the high concentration of spatial data collected by the cohort (17 students). Students want to transition from desktop GIS (ArcGIS/QGIS) to reproducible scripting in R (using `sf` and `terra`).
4. **Data Merging (Gap: 2.44) and Data Reshaping (Gap: 2.33):** Critical foundational gaps. Because environmental data is often collected from multiple sources, students cannot progress to mapping or statistics without mastering how to reshape and join tables in R.

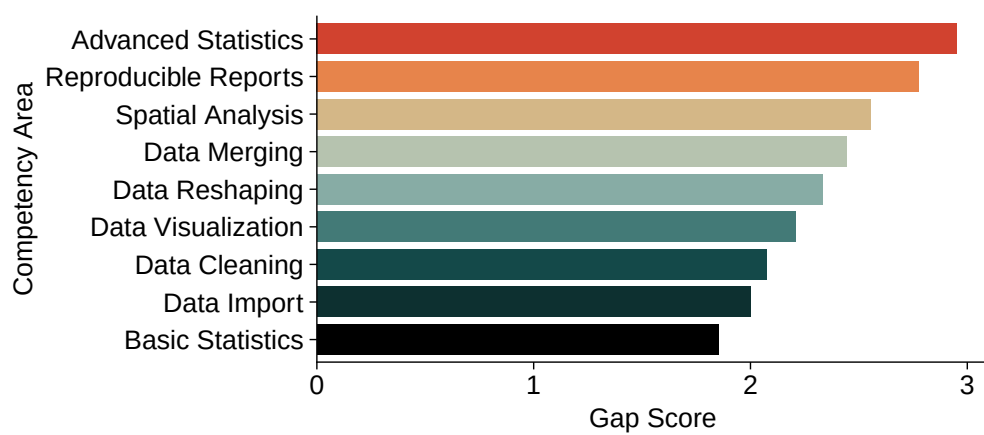


Figure 8: Identified Training Gaps by Competency

4 Proposed Training Course & Syllabus

4.1 Basic Analytical technique

To close these gaps, we propose a customized **one-week intensive course** designed specifically for environmental and climate research. The curriculum design is directly informed by the empirical findings of the TNA survey:

1. **R Rookie Baseline (Day 1):** With **77.8%** of students classified as R beginners, the course starts from the absolute basics, covering IDE setups (Positron, VS Code, or Antigravity) and syntax. The afternoon tackles the “Tidy Data” paradigm since **74.1%** are unfamiliar with it.
2. **Automating Excel Bottlenecks (Day 2):** Students spend an average of **21.1%** of their research time cleaning data. Day 2 introduces `dplyr` and `tidyr` to automate data cleaning, reshaping, and merging, directly addressing the large wrangling gaps (Data Merging: 2.44; Data Reshaping: 2.33).
3. **Publication-Ready Figures (Day 3):** Addresses the visualization gap (2.21) by replacing legacy Excel graphs with reproducible `ggplot2` charts, focusing on multi-panel faceting and journal resolution standards.
4. **Elective Track Alignment (Day 4):**
 - **Track A (Spatial GIS in R)** serves the **17** students collecting spatial data, bridging the Spatial Analysis gap (2.56).
 - **Track B (Statistical Modeling & Lab)** serves laboratory and modeling students, directly targeting the cohort’s largest overall gap: Advanced Statistics & Modeling (2.95).
5. **Automated Scientific Reporting (Day 5):** Solves the second largest gap—Reproducible Reports (2.78)—by teaching Quarto to eliminate manual document copy-pasting.

4.2 Post-Training Support: Quarterly Follow-Up Program

To ensure the long-term sustainability of the skills acquired and to provide continuous thesis-level support, we propose a structured **Quarterly Follow-Up Program** consisting of targeted **three-day sessions** (up to 3 days per quarter) over the subsequent academic year. These follow-ups will be structured dynamically to adapt to the cohort’s ongoing needs:

- **University-Specific Mentorship:** The facilitators will conduct on-site visits to the participating campuses (**SUA**, **UDSM**, and **NM-AIST**, with hybrid support for **SUZA**). These visits will focus on resolving localized database connection issues, configuring package environments in local IDEs, and providing face-to-face troubleshooting for individual student R scripts.

- **Thematic Group Deep-Dives:** Students will be grouped into their thematic clusters (e.g., Marine Ecology, GHG Inventories, Forestry, or Waste Management) for advanced code co-working. These sessions will focus on theme-specific R libraries (such as `sf/terra` for GIS, `adegenet` for genetics, or `caret` for predictive waste modeling) to solve complex analytical bottlenecks.
- **Scheduled Hybrid Progress Meetings:** Structured, scheduled progress meetings (hybrid format) where students present their current R workflows, share GitHub repositories, and receive feedback on their data visualizations and reproducible thesis draft chapters.

4.3 Course Delivery Model: “Bring Your Own Data” (BYOD)

Rather than using generic, pre-cleaned textbook datasets, the course will employ the **BYOD** approach.

1. Students will work in **Theme Tables** (4 students per table based on their research themes).
2. Afternoon laboratory sessions will be dedicated to applying the morning's lessons to their own active research datasets.
3. This guarantees immediate research progress and practical, hands-on understanding.

4.4 Syllabus

4.4.1 Day 1: Foundations of R and Tidy Data Principles

- **Morning Session (9:00 - 12:30):**
 - Cohort introductions and software setup validation.
 - Setting up reproducible project structures using modern IDE directories (such as Positron, VS Code, or Antigravity) with local folders (`/data`, `/scripts`, `/outputs`), and working with project directory instead of relative paths.
 - R syntax basics: Vectors, data frames, assigning objects, and calling functions.
- **Afternoon Session (14:00 - 17:00):**
 - Tidy Data principles (every column a variable, every row an observation).
 - Importing data into R: reading `.csv` (using `readr`) and `.xlsx` (using `readxl`).
 - Lab: Setting up research directories and importing raw thesis data.

4.4.2 Day 2: Data Wrangling and Manipulation

- **Morning Session (9:00 - 12:30):**
 - The tidyverse workflow and pipe operators (`%>%` and `|>`).
 - Core dplyr verbs: `filter()` (selecting rows), `select()` (isolating columns), `mutate()` (calculating new variables, such as carbon stock conversions), and `summarize()` combined with `group_by()` (aggregating variables).
- **Afternoon Session (14:00 - 17:00):**
 - Relational data: Joining datasets using `left_join()`, `inner_join()`, and `bind_rows()`.
 - Reshaping data: Pivoting tables between wide and long formats (`pivot_longer()` and `pivot_wider()`).
 - Lab: Merging disparate survey and environmental datasets and calculating aggregated statistics.

4.4.3 Day 3: Advanced Data Visualization with ggplot2

- **Morning Session (9:00 - 12:30):**
 - The Grammar of Graphics: Data mapping, aesthetics (`aes`), and geometries (`geom`).
 - Plot types: Scatter plots, trendlines, boxplots, and histograms.
 - Plot customization: Modifying themes, axis limits, legends, and applying color palettes (e.g., `viridis`).
 - Faceting: Creating multi-panel plots with `facet_wrap()` to display trends across sites or years.
- **Afternoon Session (14:00 - 17:00):**
 - Exporting plots: Saving publication-ready figures using `ggsave()` with precise DPI and dimension settings.
 - Lab: Creating and exporting a multi-panel, publication-quality chart using research data.

4.4.4 Day 4: Domain-Specific Elective Modules

Participants will divide into two parallel tracks based on their methodology:

- **Track A: Spatial Data & GIS in R** (For Forestry, Plastics, and Spatial GHG students)
 - Reading vector spatial data (`sf` package) and raster data (`terra` package).

- Coordinate Reference Systems (CRS) and projection management (e.g., UTM Zones for Tanzania).
- Spatial overlay, cropping rasters, and extracting raster values at coordinates.
- Mapping spatial distributions in R.
- **Track B: Statistical Modeling & Laboratory Experiments** (For Genetics, Acoustics, and Waste Lab students)
 - Hypothesis testing in R: t-tests and Analysis of Variance (ANOVA) with Tukey's post-hoc tests.
 - Simple and multiple linear regression modeling (`lm()`) and diagnostic plotting.
 - Time-series processing for laboratory runs (e.g., anaerobic digestion methane curves).
- **Joint Afternoon Session (15:30 - 17:00):**
 - Group presentations and cross-track peer feedback.

4.4.5 Day 5: Reproducible Research and Project Delivery

- **Morning Session (9:00 - 12:30):**
 - Writing reproducible research documents using Quarto (`.qmd`).
 - Combining narrative text (Markdown), code chunks, and live R tables/figures in a single file.
 - Exporting reports directly to MS Word (`.docx`), PDF, and HTML.
- **Afternoon Session (14:00 - 17:00):**
 - Final Presentations: Each Theme Table presents their end-to-end R project (data import, cleaning, visualization, and reproducible report).
 - Course evaluation, post-training questionnaire, and certificate distribution.

5 Infrastructure & Resource Requirements

To ensure a seamless training environment, the following resources must be secured prior to the course.

5.1 Hardware Requirements

1. **Laptops:** Every participant must bring a laptop with at least 8GB RAM and 40GB of free disk space.
2. **Power Outlets:** The venue must provide stable power strips and extension cables running to every participant table.
3. **Projector:** A high-definition, HDMI-compatible projector is required to display code screens clearly.
4. **Internet:** High-speed, stable Wi-Fi is mandatory, particularly on Days 1 and 4 for downloading packages and spatial libraries.

5.2 Software Prerequisites

Students must pre-install the following software using a provided installation guide sheet: * **R (v4.3.0 or higher):** [CRAN R Project](#) * **Modern IDE:** [Positron](#), [VS Code](#), or [Antigravity IDE](#) (instead of RStudio Desktop). * **Quarto CLI:** Installed as a standalone CLI or extension. * **R Packages:** tidyverse, sf, terra, here, readxl, openxlsx, vegan.

5.3 Facilitator Deliverables

- A GitHub repository containing all course slides, datasets, and Quarto templates.
- A USB flash drive pre-loaded with offline installer executables for R, modern IDEs (Positron, VS Code, or Antigravity), and a zipped package library to support students with network issues.

Appendix & Annexes

Annex A: TNA Survey Instrument (Questionnaire)

The following survey instrument was converted to an online form using KoboToolbox and sent to the 28 PhD and MSc scholarship beneficiaries.

SECTION A: Academic & Demographic Profile

1. **Full Name**
2. **Gender:** ☐ Female ☐ Male ☐ Prefer not to say
3. **Affiliated University:** ☐ Sokoine University of Agriculture (SUA) ☐ University of Dar es Salaam (UDSM) ☐ Nelson Mandela African Institution of Science and Technology (NM-AIST) ☐ State University of Zanzibar (SUZA)
4. **Level of Study:** ☐ MSc ☐ PhD
5. **Research Theme:** ☐ Coastal Forest ☐ GHG - Industrial Processes and Product Use (IPPU) ☐ GHG - Energy ☐ Marine Ecology ☐ GHG - Agriculture, Forestry, and Other Land Use (AFOLU) ☐ Plastic Waste Management ☐ GHG - Waste
6. **Thesis Title / Research Topic** (Brief draft)

SECTION B: Current Data Environment & Challenges

7. **Primary format of the data you collect/plan to collect (Select all that apply):** ☐ Tabular data (spreadsheets, CSV, database tables) ☐ Spatial data (shapefiles, GPS points, satellite rasters, DEMs) ☐ Laboratory experimental data (gas concentrations, chemical assays, molecular sequences) ☐ Survey data (household interviews, socio-economic questionnaires) ☐ Audio or sensor logs (acoustic detections, weather station outputs)
8. **Tools you currently use to store, organize, and analyze your data (Select all that apply):** ☐ Microsoft Excel ☐ Google Sheets ☐ LibreOffice Calc ☐ STATA ☐ R ☐ Python ☐ SPSS ☐ ArcGIS ☐ QGIS ☐ Other
9. **Estimated distribution of your research time (Total 100%):**
 - Data cleaning, structuring, and formatting: _____%
 - Data analysis, statistics, and modeling: _____%
 - Creating figures, charts, and maps: _____%
 - Report, thesis, or paper writing: _____%
10. **What is your single biggest bottleneck or frustration when managing your research data?**

SECTION C: Programming & R Experience

11. **Have you ever written code in R or RStudio before?** ☐ No, I have never used R. ☐ Yes, I have run scripts written by others, but cannot write my own from scratch. ☐ Yes, I can write basic scripts to load and analyze data. ☐ Yes, I write custom R scripts regularly and feel comfortable using packages.
12. **Are you familiar with the concept of “Tidy Data”?** ☐ Yes ☐ No
13. **Are you familiar with version control tools like Git and GitHub?** ☐ Yes ☐ No

SECTION D: Core Competencies Self-Assessment

Rate your current skill level (1 = None, 5 = Expert) and your interest in learning (1 = Not interested, 5 = Crucial):

Competency Area	Current Skill	Interest in Learning
Data Import: Loading CSV, Excel, TXT, or spatial files into R	1 - 2 - 3 - 4 - 5	1 - 2 - 3 - 4 - 5
Data Cleaning: Filtering rows, selecting columns, renaming variables, handling NAs	1 - 2 - 3 - 4 - 5	1 - 2 - 3 - 4 - 5
Data Reshaping: Pivoting tables (wide <-> long), splitting strings	1 - 2 - 3 - 4 - 5	1 - 2 - 3 - 4 - 5
Data Merging: Joining different datasets using common keys	1 - 2 - 3 - 4 - 5	1 - 2 - 3 - 4 - 5
Data Visualization: Creating publication-quality plots using ggplot2	1 - 2 - 3 - 4 - 5	1 - 2 - 3 - 4 - 5
Spatial Analysis: Reading, plotting, and manipulating coordinates or GIS shapefiles	1 - 2 - 3 - 4 - 5	1 - 2 - 3 - 4 - 5

Competency Area	Current Skill	Interest in Learning
Basic Statistics: Running t-tests, ANOVA, linear regressions, model parameters	1 - 2 - 3 - 4 - 5	1 - 2 - 3 - 4 - 5
Advanced Statistics/ML: Predictive models, time-series, clustering, ML	1 - 2 - 3 - 4 - 5	1 - 2 - 3 - 4 - 5
Reproducible Reports: Creating Word/PDF documents directly using Quarto	1 - 2 - 3 - 4 - 5	1 - 2 - 3 - 4 - 5

SECTION E: Expectations & Comments

14. **What specific research question do you hope this training will help you answer using R?**
15. **Any other suggestions or requirements for the training facilitators?**

Annex B: Cohort Registry and Responses Summary

The table below lists the 27 students who responded to the survey, including their university affiliation, level of study, and research theme.

Table 7: IUCN Scholarship Beneficiaries Survey Registry

Student Name	University	Level	Research Theme
Imakulata Raphael Mwenda	NMAIST	MSc	Waste
Musa Yusuph Mungah	NMAIST	MSc	Coastal Forest
SITTA MALULU BULUMA	NMAIST	MSc	Coastal Forest
Bhoke Masisi	NMAIST	PhD	Ippu
Dativa Byarufu	NMAIST	PhD	Waste
ALSELM BONAVENTURE SENGGERERE	SUA	MSc	Plastic Waste
Aikande Frank Lema	SUA	MSc	Ippu
Bahati Beatrice Pius	SUA	MSc	Energy
EMANUEL H LENGEJU	SUA	MSc	Energy
ESTER ELEUTERI TEMBA	SUA	MSc	Coastal Forest
Faraja Saulu Kakulwa	SUA	MSc	Marine Ecology
GIVEN EDWARD MWAKYEMBE	SUA	MSc	Afolu
Gladness Edwin Mguluka	SUA	MSc	Energy
JOEL SAMWEL MANDIA	SUA	MSc	Ippu
Musa Luhula Bujunju	SUA	MSc	Afolu
Perpetua Nyagwaswa	SUA	MSc	Ippu
gladness edwin mguluka	SUA	MSc	Energy
Diana Lawrence Tesha	SUA	PhD	Coastal Forest
Sarafina Masanja	SUA	PhD	Energy
Maryam Ramadhan Mwinyi	SUZA	MSc	Plastic Waste
ANNETTE AMBROSE KESSY	UDSM	MSc	Plastic Waste
MWAJABU FADHILI MKINDI	UDSM	MSc	Waste
Mustapha Issa	UDSM	MSc	Marine Ecology
Stanley Wilson Charles	UDSM	MSc	Waste
David Kavana	UDSM	PhD	Marine Ecology
Lynder Gesase	UDSM	PhD	Plastic Waste
Lynder Gesase	UDSM	PhD	Plastic Waste

Annex C: Zipped Package Setup Script

The following R script can be distributed to participants to install all necessary CRAN and Bioconductor packages before starting the course.

```
# =====
# BAHARI YETU DATA MANAGEMENT TRAINING PACKAGE INSTALLER
# =====
# Run this script in your IDE (Positron, VS Code, or Antigravity) to install all
# packages required for the training course.

required_packages <- c(
  "tidyverse", # Core suite (dplyr, ggplot2, readr, tidyr, purrr, stringr)
  "sf",        # Vector spatial data processing
  "terra",     # Raster spatial data processing
  "here",      # Relative path management
  "readxl",    # Reading Excel spreadsheets
  "openxlsx",  # Exporting multi-sheet Excel workbooks
  "vegan",     # Vegetation and community ecology indices
  "cluster",   # Phytochemical profiling and clustering
  "randomForest", # Machine learning predictions for waste/GHG models
  "caret",     # Machine learning regression/classification framework
  "knitr",     # Table generation and dynamic document compiling
  "rmarkdown", # Markdown document rendering support
  "scales"     # Visual scale styling support
)

# Identify missing packages
missing_packages <- required_packages[!(required_packages %in% installed.packages()[, "package"])]

if (length(missing_packages) == 0) {
  message("All required packages are already installed!")
} else {
  message("Installing the following missing packages: ", paste(missing_packages, collapse = ", "))
  install.packages(missing_packages, repos = "https://cloud.r-project.org")
  message("Package installation completed!")
}
```



Bahari Yetu Project

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